

Unreal Engine 3 / Unreal Tournament 3 Package File Specification (Reference Guide)

Compiled from multiple community and reverse-engineering sources. Intended for engineers extending UE1/UE2 readers to UE3 (.upk, .umap, .u, etc.).

Overview

UE3 continues the Unreal Package system from UE1/UE2 with changes to headers, string encodings, compression, and table formats.

This reference consolidates information from Gildor's UE Viewer, EliotVU's UELib, BeyondUnreal wiki, and Epic community notes.

High-Level Layout

1. Header (file summary + generations)
2. Tables (Name, Import, Export)
3. Object data (serialized, often chunk-compressed)

The layout conceptually matches UE1/UE2, but UE3 introduces compression flags and 32-bit string/flag widths.

Signature and Versioning

- Signature: 0x9E2A83C1 (same as UE1/UE2)
- FileVersion: 32-bit value (low 16 bits = PackageVersion, high 16 bits = LicenseeVersion)
- Version examples: ~512 for UT3, later UDKs ~800+
- Always split version into two 16-bit components for compatibility.

Header Fields (FPackageFileSummary approximation)

Common UE3 header fields (in order):

- Tag (0x9E2A83C1)
- FileVersion (PackageVersion + LicenseeVersion)
- HeaderSize
- PackageGroup / FolderName (string)
- PackageFlags (u32)
- NameCount / NameOffset
- ExportCount / ExportOffset
- ImportCount / ImportOffset
- DependsOffset (exists when PackageVersion $\geq$ 415)
- GUID (16 bytes)
- Generations (count + repeated struct)

Use these offsets to locate tables; do not assume the header ends before them.

String Encoding (FString)

UE3 replaces ASCIIZ/byte-length strings with FString:

- int32 length prefix

>0 → ANSI/UTF-8 (includes NUL)

<0 → UTF-16LE (abs(length) code units, includes NUL)

=0 → empty string

Always trim the trailing NUL after reading. Strings appear in names, imports, exports, and properties.

Name Table

Each entry: FString + Flags (64-bit preferred).

Count/Offset come from header. Names remain unique, include "None" sentinel.

For older transitional packages, Flags may still be 32-bit. Probe high dword for zeros to confirm.

Object References

Encoding rules persist:

- 0 = null reference
- <0 = import (−Index−1)
- >0 = export (Index−1)

UE3 typically stores references as plain 32-bit integers rather than CompactIndex, but both may appear.

Import Table

Fields:

- ClassPackage (Name idx)
- ClassName (Name idx)
- OuterIndex (obj ref)
- ObjectName (Name idx)

Same structure as UE1/UE2, but all indices are often i32 in UE3.

Export Table

Fields:

- ClassIndex, SuperIndex, PackageIndex (obj refs)
- ObjectName (Name idx)
- ObjectFlags (usually u64)
- SerialSize (i32 or CompactIndex)
- SerialOffset (i32) if SerialSize > 0

All offsets and sizes are absolute positions within the file (or decompressed stream).

Compression and Cooked Packages

UE3 adds package-level compression for cooked builds:

- Header includes CompressionFlags and a CompressedChunks table.
- Each chunk has compressed/uncompressed sizes.
- To read, decompress all chunks sequentially into a virtual buffer, then seek via that buffer.

Compression often uses Zlib or LZO algorithms (per platform).

Never seek raw offsets when compression is active.

Depends Table

If PackageVersion ≥ 415 , the header includes DependsOffset.

This table lists dependencies for exports to control load order. Non-essential for basic parsing.

Properties (Changes from UE1/UE2)

- Same info byte (type, size, array flag).
- BoolProperty value still encoded in info byte bit 4 (0x10).
- StringProperty payloads now use FString (signed length prefix).
- Struct/Vector/Rotator structures unchanged in concept (12-byte numeric payloads).

Common Field Widths

- ObjectFlags $\rightarrow$ 64-bit
- PackageFlags $\rightarrow$ 32-bit
- Indices $\rightarrow$ 32-bit (signed)
- Names $\rightarrow$ FString + 64-bit flags

Backward compatibility:

- CompactIndex still appears in some early UE3 builds.
- Maintain your CompactIndex decoder as a fallback.

Typical Version Thresholds

UE2: ~100–200

UE3: ~334–800

UE3 introduces:

- 334+ $\rightarrow$ compression chunks
- 415+ $\rightarrow$ DependsOffset
- 512+ $\rightarrow$ UT3 baseline

Tools and Reference Implementations

- UE Viewer (Gildor) — <https://www.gildor.org/en/projects/umodel>
- UELib by EliotVU — <https://github.com/EliotVU/UELib>
- BeyondUnreal Wiki — <https://beyondunrealwiki.github.io/pages/package-file-format.html>
- Epic Engine source references (FString, serialization) — UDK headers and code comments.

Implementation Notes for PHP Reader Extension

1. Add a branch for \$version ≥ 334 to switch to UE3 mode.
2. Use i32 string lengths (FString).
3. Use u64 flags for names and objects.
4. Prefer plain i32 indices over CompactIndex (keep fallback).
5. When CompressionFlags $\neq 0$, read via uncompressed buffer.
6. For BoolProperty, use info-bit 0x10 for value.
7. For StringProperty payloads, call FString reader.
8. Handle DependsOffset optionally; not needed for content parsing.

Summary

UE3 packages preserve the Unreal Package structure but adopt 32-bit widths, FString encoding, and compression.

With the above adjustments, a UE1/UE2 reader can be extended for UT3/UE3-era games with minimal structural change.

Primary References and Credits

- 1. Gildor (UE Viewer) forums — UE3 package header and compression discussion.
- 2. EliotVU (UELib) GitHub — comprehensive C# library for UE3 packages.
- 3. BeyondUnreal Wiki — Unreal Package File Format overview.
- 4. Epic documentation and UDK source snippets (FString, serialization).
- 5. Reverse-engineering notes across community projects (UT3, UDK, UE3).